

Zhongfang He

(PhD-Economics, FRM)

Contact Information

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Education

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| 2009 | PhD, Economics, University of Toronto, Canada (2005-2009), specializing in econometrics, thesis “ Structural Breaks and Forecasting in Empirical Finance and Macroeconomics ”. |
| 2005 | MA, Economics (doctoral stream with full scholarship), University of Toronto, Canada (2004-2005) |
| 2004 | BA, Economics, Nankai University, China (2000-2004), top graduation grade |

Working Experience

- 2026 - Present: Senior Manager, Advanced Analytics and Modeling (highest non-executive level), TD Bank.
- Develop models for enterprise capital stress testing of the investment banking/trading segment of TD bank.
 - Build the machine learning capacity for predicting revenues and balance sheets.
- 2017 - 2026: Director, Quantitative Analytics, Royal Bank of Canada.
- Manage the quantitative analytics function for banking-book/non-trading market risk and asset liability management.
 - Pioneer the use of machine learning and AI for prediction and risk modeling of financial products.
 - Develop models for interest rate risk management (build statistical models to predict client behaviors, price and hedge the option risk embedded in banking products, stress test and scenario analysis, etc.).
 - Analyze quantitative strategies to improve hedging efficiency.
 - Help develop models and analytics for liquidity risk management and funds transfer pricing.
- 2011 – 2017: Senior Manager, Asset Liability Management and Portfolio Analytics, Royal Bank of Canada.
- Develop models for interest rate risk management, including deposits, loans and mortgage commitments etc.
 - Develop models for liquidity risk management, including deposits, credit lines and credit cards etc.
 - Develop models for stress tests including enterprise-wise stress test, OSFI-Bank of Canada Macro stress test, and US Fed CCAR.

2011: Senior Analyst, Risk Analytics, Manulife Financial.

- Develop assumptions for long-run government bond yields, corporate spreads and equity returns of different countries.
- Provide recommendations to the development of structural break models for equity returns.
- Participate in the insurance industry working group of developing capital requirement models.

2009 – 2011: Senior Analyst, Financial Sector Specialist, Bank of Canada.

- Develop model for separating the credit and liquidity risks of corporate bond spreads.
- Develop model for measuring the effect of unconventional monetary policy on Canadian market interest rates.
- Develop model for extracting market expectation of monetary policy rate via overnight index swap rate.
- Model the credit, funding liquidity and contagion risks of Canadian financial institutions.
- Evaluate the predictive power of the yield curve for subsequent macroeconomic performance.

Professional Designation

Financial Risk Manager (FRM), 2010

Computer Programming Skills

Python, Matlab, C/C++, R, SQL, SAS

Published Research Work

“[Locally Time-Varying Parameter Regressions](#)”, 2024, *Econometric Reviews*, 43(5), pages 269-300.

“[Time Dependent Shrinkage of Time-Varying Parameter Regression Models](#)”, 2024, *Econometric Reviews*, 43(1), pages 1-29.

“[A Dynamic Binary Probit Model with Time-Varying Parameters and Shrinkage Prior](#)”, 2023, *Journal of Business and Economic Statistics*, 42(1), pages 335-356.

“[A Computationally Efficient Mixture Innovation Model for Time-Varying Parameter Regressions](#)”, 2023, *Econometrics and Statistics*, in press.

“[Integrating Macroeconomic Variables with Interest Rate Scenarios for Interest Rate Risk Measurement in the Banking Book](#)”, 2020, *Journal of Risk*, 22(5), pages 67-82.

“[Real Time Detection of Structural Breaks in GARCH Models](#)”, joint with John Maheu, 2010, *Computational Statistics and Data Analysis*, 54(11), page 2628-2640.

Other Research Work

“[Understanding Systemic Risk: The Trade-Offs between Capital, Short-Term Funding and Liquid Asset Holdings](#)”, joint with Celine Gauthier and Moez Souissi, 2010, presented at the workshop of the Bank of Canada.

[“Evaluating the Effect of the Bank of Canada’s Conditional Commitment Policy”](#), 2010, presented at the workshop of the Bank of Canada.

[“Monetary Policy and Time Varying Risk Premium: Evidence from the Canadian Overnight Index Swap Market”](#), joint with Timothy Grieder, 2010, presented at the workshop of the Bank of Canada.

[“Credit and Liquidity Risks in Canadian Corporate Spreads: A Latent Factor Approach”](#), joint with Jean-Sebastien Fontaine and Timothy Grieder, 2010, presented at the workshop of the Bank of Canada.

[“Forecasting Output Growth by the Yield Curve: the Role of Structural Breaks”](#), presented at the 2009 Annual Conference of the Canadian Economic Association and the workshop of the Bank of Canada.

[“A Tilt Stochastic Volatility Model with Leverage Effect”](#), presented at the 2007 International Conference on Computing in Economics and Finance.

Other Professional Experiences

2004-2008: Teaching Assistant, University of Toronto

- Graduate courses: econometrics, financial econometrics,
- Undergraduate courses: quantitative methods in economics, macroeconomics, microeconomics.

2006-2008: Research Assistant, University of Toronto

- Participated in various research projects on financial asset pricing.